



03 February 2025

Subject: Response to Request for Information on AI Action Plan

To: Faisal D'Souza

National Coordination Office, NITRD

2415 Eisenhower Avenue, Alexandria, VA 22314, USA

Dear Mr. D'Souza,

We appreciate the opportunity to provide input on the development of the AI Action Plan as outlined in Executive Order 14179. As a private engineering company actively engaged in AI applications, we have worked on projects that leverage AI for infrastructure evaluation, asset management, financial modeling, and large-scale sensor data processing. Our experience has shown that AI is a critical tool in solving complex challenges across multiple industries, from improving public infrastructure planning to enhancing financial market efficiency.

The development of AI-driven solutions in these key areas is essential for national economic growth, sustainability, and technological leadership. AI's ability to synthesize vast amounts of data, automate decision-making, and identify emerging trends provides significant value across multiple domains. Below, we outline our perspectives and recommendations on AI policy actions that will foster innovation and responsible AI adoption.

1. AI for National-Level Infrastructure Evaluation and Forecasting

Efficient and data-driven evaluation of national infrastructure, particularly roads and bridges, is critical for maintenance planning and budget forecasting. AI has the capability to:

- Automate condition surveys using machine learning models trained on historical data, satellite imagery, and field inspections.
- Develop predictive models for infrastructure deterioration, supporting optimized rehabilitation budgets and long-term asset management strategies.
- Enhance integration with existing GIS and digital twin technologies, enabling dynamic, real-time analysis for state and local government agencies.

Recommended Policy Actions:

- Encourage the standardization and accessibility of infrastructure condition datasets to facilitate AI training and model development.
- Support pilot programs integrating AI into national and state-level Department of Transportation (DOT) planning processes.
- Promote partnerships between government agencies, engineering firms, and AI developers to enhance AI-driven decision support systems for public infrastructure.

2. AI in Electrical Network Asset Inventory and Identification

The accurate inventory of electrical network assets is fundamental to efficient grid management.

AI applications in this area include:

- Automated identification of assets from field photographs using computer vision models.
- Machine-readable identification sign planning, optimizing asset tracking and maintenance logistics.
- AI-powered geospatial analysis to map electrical infrastructure and predict maintenance needs.

Recommended Policy Actions:

- Establish national guidelines for AI-based asset recognition and tagging in power infrastructure.
- Facilitate data-sharing mechanisms between utilities, regulatory bodies, and AI developers for improved asset inventory management.
- Develop cybersecurity standards to ensure the integrity and confidentiality of AI-driven asset identification processes.

3. AI for Large Sensor Network Interpretation and Processing

The increasing deployment of sensor networks—such as strain gauges, accelerometers on bridges, and weight sensors on roads—presents an opportunity for AI-driven real-time data analysis. AI applications can:

- Identify early warning signs of structural deterioration, improving safety and reducing long-term maintenance costs.

- Optimize sensor data interpretation, distinguishing between routine environmental effects and critical structural issues.
- Enable smart infrastructure monitoring, allowing for automated reporting and integration with predictive maintenance platforms.

Recommended Policy Actions:

- Support AI research initiatives focused on processing large-scale sensor network data for infrastructure health monitoring.
- Standardize sensor data formats to facilitate AI model training and interoperability.
- Invest in AI-powered predictive analytics platforms for national infrastructure resilience programs.

4. AI in Financial Markets and Investment Strategies

Beyond engineering and infrastructure applications, AI has the potential to transform financial markets through advanced trading strategies and fund management. Key applications include:

- AI-driven trading algorithms that analyze large datasets to identify profitable investment opportunities.
- Risk management models leveraging AI for portfolio optimization, stress testing, and anomaly detection.
- Market trend prediction using deep learning and natural language processing (NLP) to assess economic indicators, investor sentiment, and financial news.

Recommended Policy Actions:

- Support regulatory frameworks that balance AI innovation in trading with financial stability and transparency.
- Encourage research into AI-based financial modeling to improve market efficiency and risk mitigation.
- Promote AI-driven investment funds with responsible governance and compliance mechanisms to enhance financial market resilience.

5. Navigating Copyright Law in AI Model Training and Standards Development

As AI adoption grows, considerations surrounding copyright law and fair use are essential for training AI models responsibly. Beyond data privacy and intellectual property concerns, AI plays a critical role in the drafting, refinement, and standardization of engineering and technical standards. Key considerations include:

- **AI for Writing Standards:** AI can assist in drafting and refining engineering, safety, and regulatory standards, ensuring consistency and clarity while reducing ambiguities.
- **Error Reduction and Quality Control:** AI can analyze historical revisions of standards to identify common errors, inconsistencies, and potential conflicts across various regulations, reducing the risk of misinterpretation.
- **Synthesis and Standardization of Approaches:** AI can evaluate diverse global standards on a given topic, identify best practices, and propose harmonized approaches, facilitating greater interoperability and alignment across industries.

Recommended Policy Actions:

- Establish clear legal frameworks distinguishing permissible data use under fair use exceptions versus requiring explicit permissions, particularly for AI models processing technical standards and regulatory documents.
- Encourage AI-assisted drafting and quality control in technical standards development to improve consistency and reduce conflicts.
- Support open-access initiatives for technical and engineering standards, allowing AI models to process and synthesize best practices across regulatory frameworks.

6. AI in Digital Twin Development for Infrastructure Preservation and Monitoring

Digital twins—high-fidelity virtual models of physical assets—are becoming essential for the long-term preservation, monitoring, and engineering analysis of critical infrastructure. AI enhances the development and application of digital twins by enabling real-time data processing, predictive analytics, and automated decision support.

Key Applications of AI in Digital Twins for Bridges and Infrastructure:

- **Preservation Planning for Historic Bridges:** AI-powered digital twins can be used to model aging infrastructure, simulate deterioration over time, and recommend preservation strategies while maintaining historical integrity.

- **Structural Health Monitoring:** AI-driven analysis of sensor data (e.g., strain gauges, accelerometers) within digital twins helps detect anomalies, assess fatigue, and predict potential failures.
- **Planning Retrofitting and Upgrades:** Engineers can use AI-enhanced digital twins to simulate the effects of retrofits, seismic reinforcements, and load changes before implementing costly physical modifications.
- **Long-Term Monitoring and Asset Management:** AI continuously refines digital twin models based on new data, improving long-term asset management strategies for state and federal transportation authorities.

Recommended Policy Actions:

- Promote federal and state investment in AI-driven digital twins for key infrastructure assets, particularly aging and historically significant bridges.
- Encourage standardized data collection protocols to enhance interoperability between AI models and digital twin platforms.
- Support research and pilot projects demonstrating AI's role in improving infrastructure lifecycle management through digital twins.

Conclusion

As AI continues to evolve, its applications in infrastructure evaluation, asset inventory, sensor data analysis, and financial markets offer transformative benefits. To maximize impact, policies should focus on:

- Data standardization and accessibility to improve AI model training and real-world applicability.
- Industry-government collaboration to drive AI adoption in critical infrastructure sectors.
- Investment in AI-driven monitoring and forecasting to support data-informed decision-making.
- Legal clarity on AI model training and copyright considerations, particularly for AI's role in drafting and refining technical standards to ensure consistency, accuracy, and harmonization across industries.
- Encouraging AI in financial innovation while ensuring compliance, stability, and responsible governance.

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We appreciate your consideration of these recommendations and welcome the opportunity for further discussion or collaboration on AI applications in infrastructure, asset management, and financial markets.

Sincerely,

Justine Marik Touma

Chairman

Engineering & Research International

John Touma

Manager

Engineering & Research International